How to Use the Unity Projection System

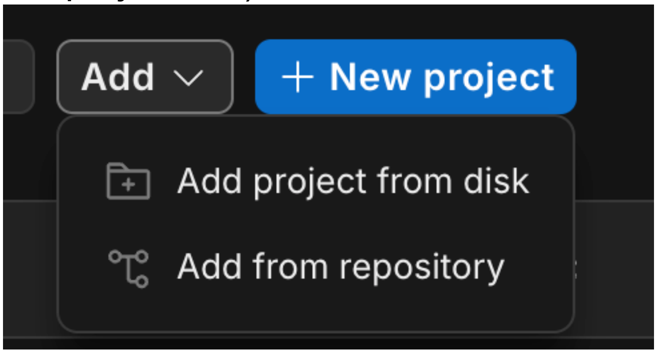
1. Project Information

- Name: Project_training2025_UNIVPM_eng_techviz test2.6-2
- Location: /Users/rwth/Documents/3-FanYe/Model_Italy/Italy_Model/Project_training2025_UNIVPM_eng_techviz test2.6-2
- Unity version:2020.1.17f1

Before modifying the project, create a backup of the complete project folder.

2. Opening the Project

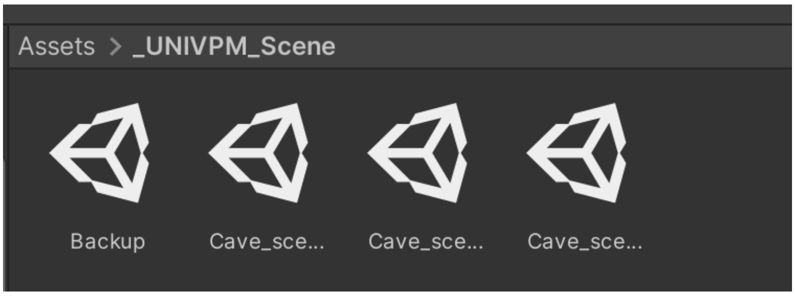
1.) Open Unity Hub.
2.) Select Add project from disk. (Alternatively, if the project is already listed in Unity Hub, simply open it directly from the project list.)



3.) Select the project folder in the correct location with the correct name containing:
  - Assets
  - Packages
  - Project Settings

4.) Open the project with the correct Unity version.

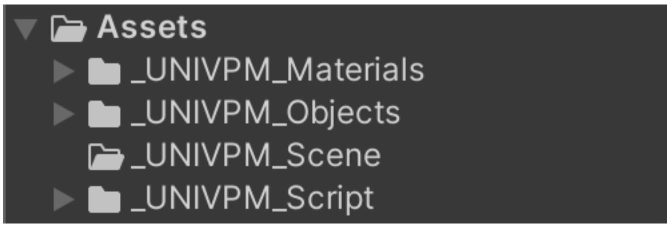
5.) Open the required scene from Assets/Scenes.



3. Project Structure

The most important folders are:

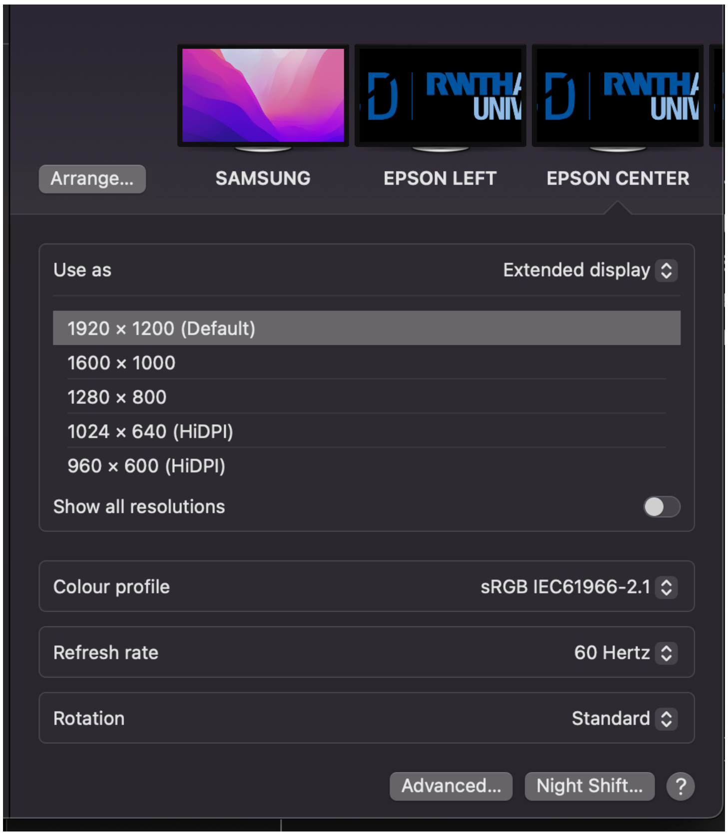
- UNIVPM_Scene: Unity scenes;
- UNIVPM_Script: C# control scripts;
- UNIVPM_Materials: colours and surface materials;
- UNIVPM_Objects: reusable objects;



4. Starting the Projection

Before starting Unity:

- Switch on all three projectors.
- Connect them to the Mac.
- Open System Settings → Displays.
- Check the display arrangement.
- Set the projector resolution to 1920 × 1200



- Open Qlab
- Settings confirm

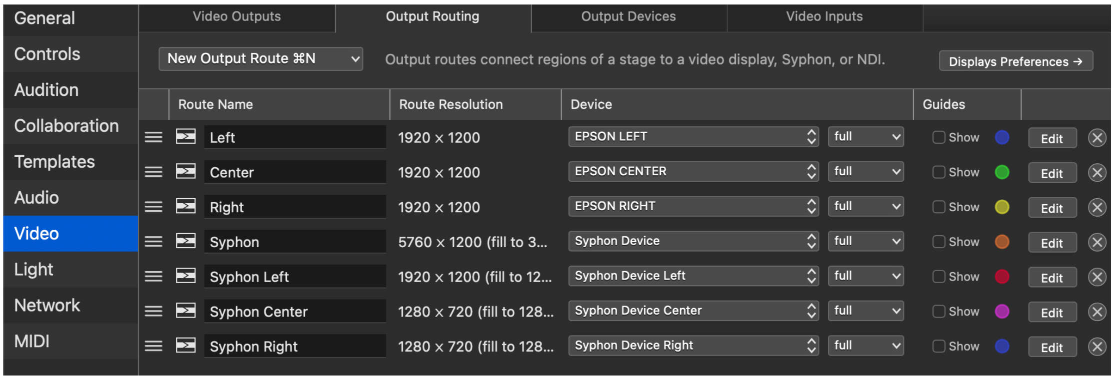
The system uses three Devices:

Device	Projection wall
EPSON FRONT	Front wall
EPSON LEFT	Left wall
EPSON RIGHT	Right wall

Each device must use the correct **Target Display**.

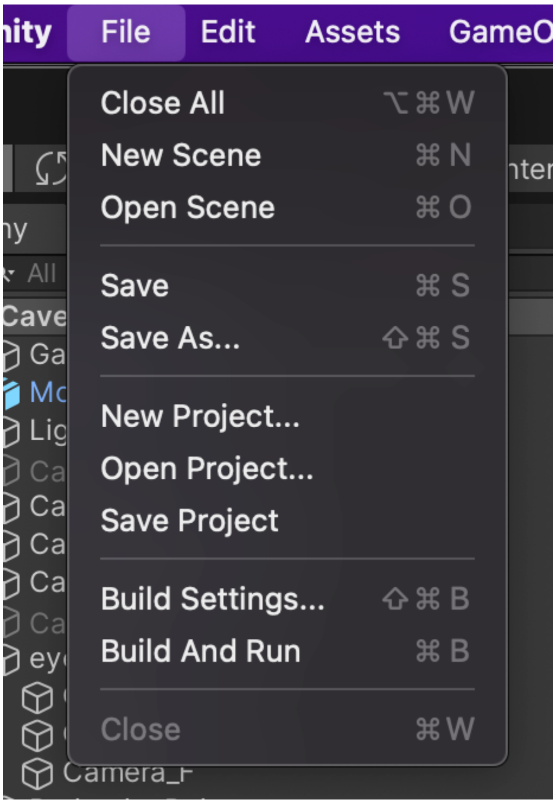
Use the following path to configure the correct output.

File → WorkSpace Settings → Video → Output Routing



- Select **"Projector On"** with the mouse and press the **Spacebar**.

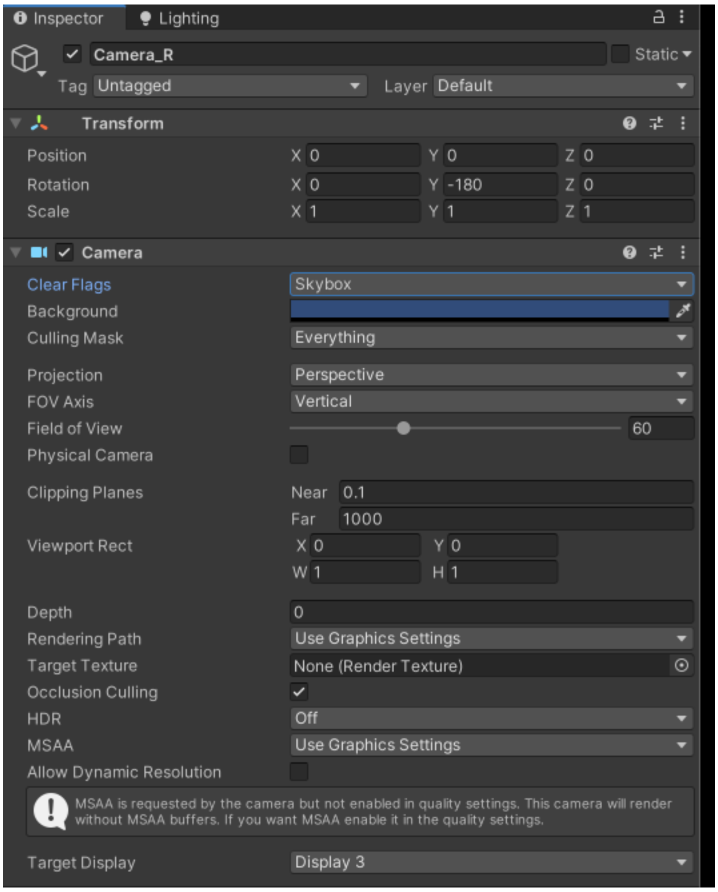
- Open Unity, open *File* → *Build and Run* to verify that all three projection walls display the correct content.



- Press **Command + Q** to exit this mode.

- If not, adjust the Target Display setting of each of the three cameras under eyeroot.

- Select the Camera_R, Camera_L and Camera_F under eyeroot in the Hierarchy.
- Change Target Display in the Inspector.
- Assign **Displays 2, 3, and 4** to Cam_L, Cam_F, and Cam_R in the correct combination.
- HDR **must off**. Incorrect settings may cause screen flickering.



- The eyeroot settings must be adjusted separately in every scene.

- The display should be checked before each experiment because it may change after restarting the computer.

5. Camera and eyeroot Settings

The current camera orientations are approximately:

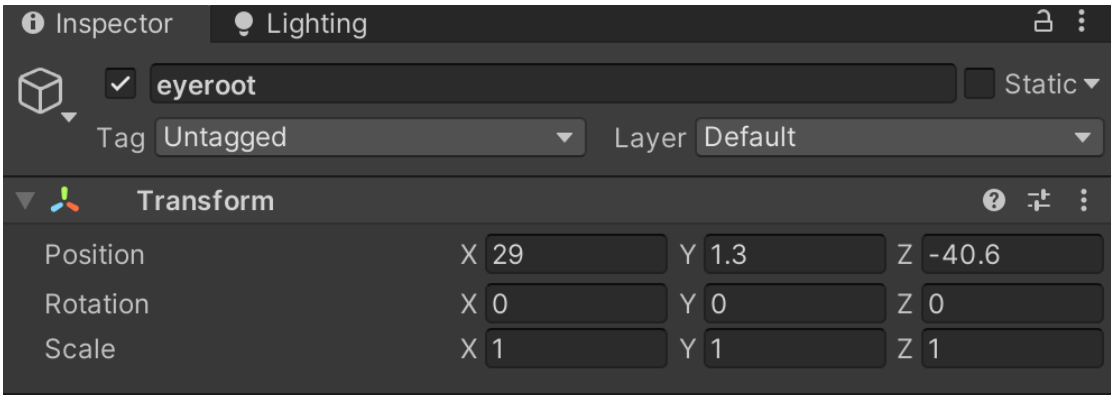
Camera	Y Rotation
Cam_F	+90°
Cam_L	0°
Cam_R	-180°

The reference observer's eye position is:
Eyeroot = (x(blue), y(green), -z(red))
Normally, **Eyeroot = (29, 1.3, -40.6)**

In general, the x and z coordinates of the eyeroot remain fixed, while the y coordinate is adjusted according to the observer's eye height.

To adjust the Eyeroot:

- Select the **eyeroot** in the Hierarchy.
- Change its X, Y, or Z position in the Inspector.
- Start Play Mode.
- Check whether the images align correctly across the three walls.



6. Off-Axis Projection

The script `OffAxisProjection` calculates the correct perspective for each projection wall. The observer's eye position corresponds to the **eyeroot** in Unity. The physical corner points of each projection wall correspond to the corner points defined in **ProjectionPoints** within the Hierarchy. Therefore, **ProjectionPoints should not be changed**. The position of the EyeRoot is determined by the relative position between the projection walls and the observer's eyes.

It uses:

- the observer's Eye position;
- the three corner points of the projection wall;
- the camera assigned to that wall.

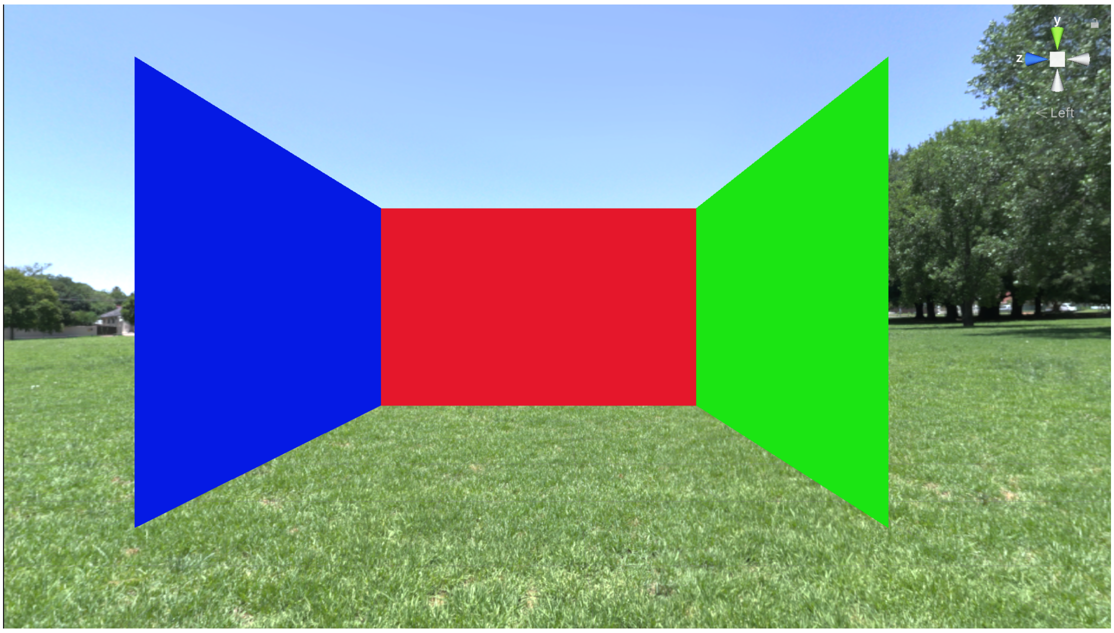
The corner points are normally:

- pa: lower-left corner;
- pb: lower-right corner;
- pc: upper-left corner.

The projection walls defined by their corner points are positioned as follows:

- Front wall: x = 32, y = 0.2 to 2.7, z = -38.6 to -42.6
- Left wall: x = 28 to 32, y = 0.2 to 2.7, z = -38.6
- Right wall: x = 28 to 32, y = 0.2 to 2.7, z = -42.6

For illustration purposes, the figure below shows the physical arrangement of the projection walls. The projection walls themselves are **not included** in the actual Unity model.



Incorrect settings may cause distorted images, black screens, or projection matrix calculation errors.

The camera Field of View should not normally be adjusted manually because it is calculated by the script.

7. Main Scripts

- **Projection Control Scripts**

Calculates the asymmetric projection matrix for the three-wall projection system.

Script list: `OffAxisProjection`

- **Scene Management Scripts**

These scripts control transitions between Scene 0, Scene 1, and Scene 2.

Script list: `SceneLoader_NextScene`

- **Scene 0 UI Control Scripts**

These scripts control the UI elements and interactions used in Scene 0.

Script list: `Scene0_Survey`

- **Scene 1 UI Control Scripts**

These scripts control the UI elements and interactions used in Scene 1, including the display of energy costs, thermal comfort, air quality, the status of the heating, window, and blinds, as well as the tips shown on the TV screen. It can be found in the _UNIVPM_Script/TextMeshPro folder.

Script list: `RandomTipsOnTV`, `Scene1_StatusAndComfort`

- **Scene 2 UI Control Scripts**

These scripts control the UI elements and interactions used in Scene 2.

Script list: `Scene2_Survey`

8. Modifying Scene Content

- **Light**

  - In the Hierarchy, there is a Lights object containing five different types of lights. Select the light you want to adjust and modify the relevant parameters in the Inspector.

- **Move, Rotate, or Scale an Object**

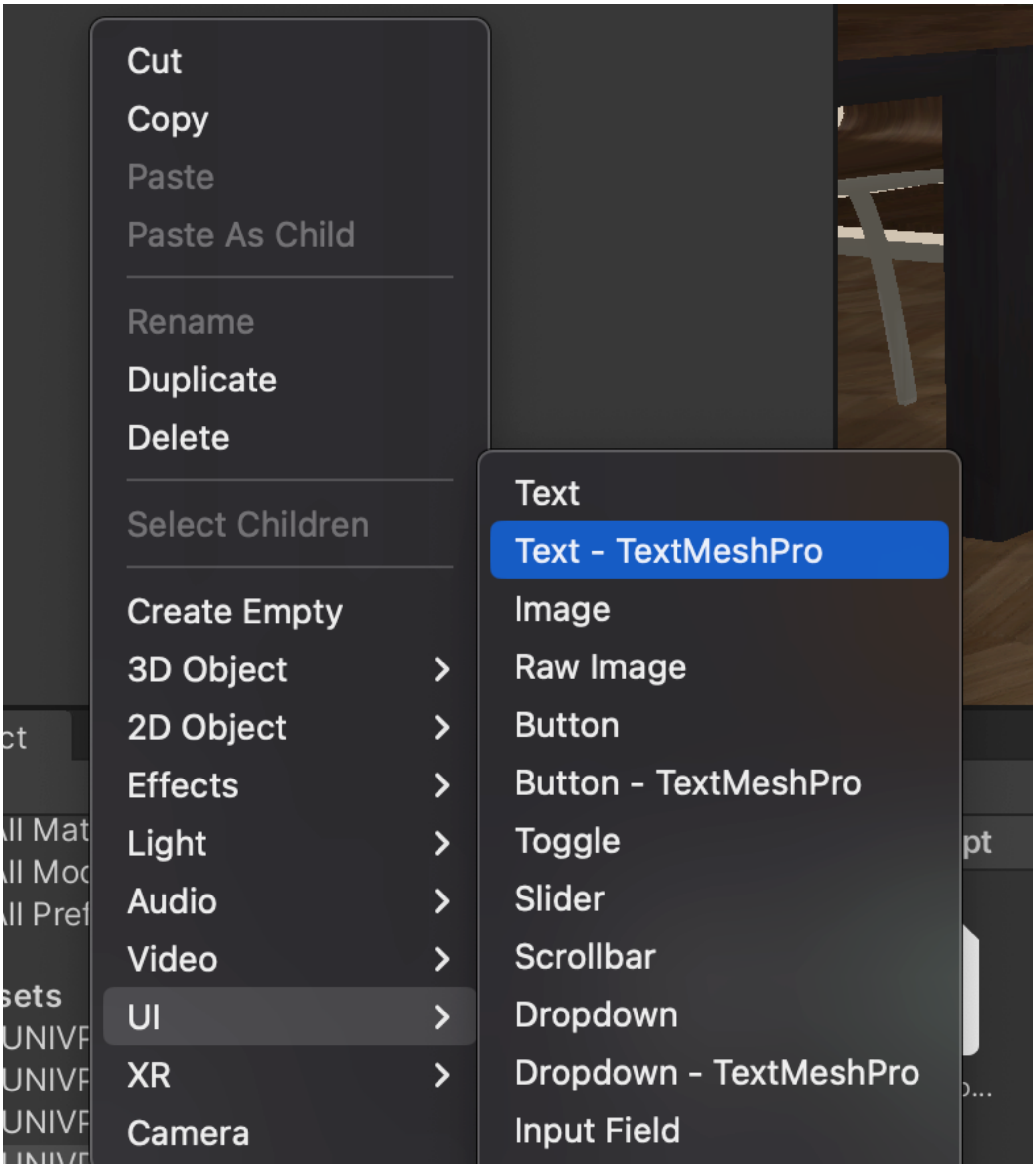
  - change the Transform values in the Inspector.

- **Replace an Image**

  - Import the new image into Assets.
  - Select the relevant UI object.
  - Replace the Source Image in the Inspector

- **Change the text**

In Hierarchy create *UI-text-TextMeshPro* object and use it together with the corresponding script.



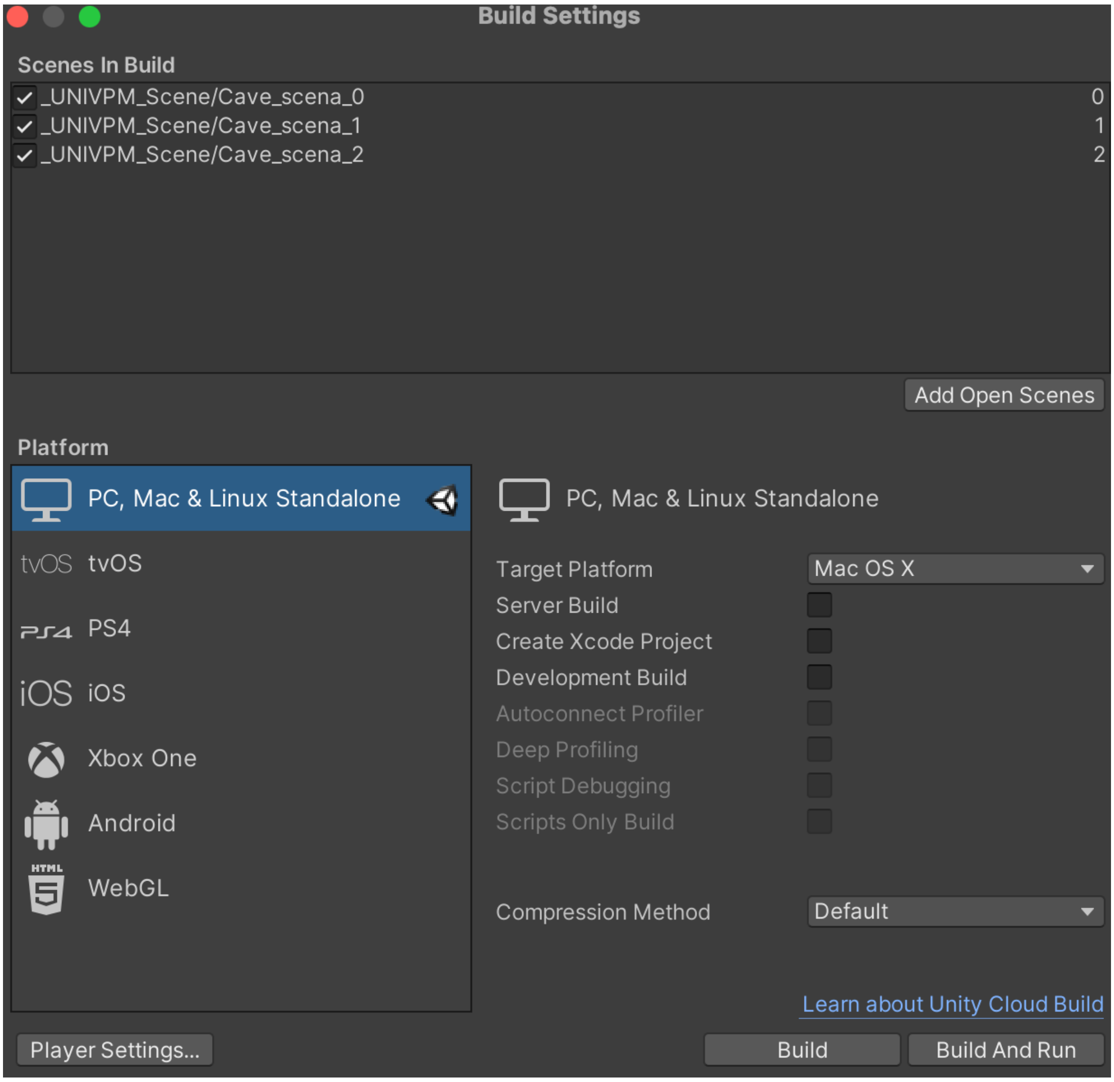
- **Change a Material**

  - Select the object.
  - Find the assigned material in the Mesh Renderer.
  - Change the colour or texture.

A shared material affects all objects using that material.

9. Creating a new build

- Open File → Build Settings
- Select Platform “PC,MAC and Linux Standalone”.
- Add Scene 0, Scene 1, and Scene 2.
- Check the scene order.
- Select Build and Run to create the build and automatically launch it on the connected machine.
- If you only need to generate the application without launching it, select Build instead.
- Save the build **outside the Assets** folder.



10. Operating Instructions After Launching the Program

- In Scene 0 and Scene 2, use the Up Arrow and Down Arrow keys to switch between questionnaire items.

- **In Scene 1, use:**
  - H to control the heating.
  - W to control the window.
  - B to control the blinds.

- Use the Right Arrow key to switch to the next scene. **Do not use the Left Arrow key.**

11. Notes and Tips

- **Changes made during Play Mode are not saved**

Always make permanent changes while not in Play Mode.

- **Back up the project before major changes**

This makes it possible to restore a working version if something goes wrong.

- **Save carefully**

It is often easier to spend a few minutes checking changes than to undo mistakes later.

- **Test after making changes**

Small changes can sometimes affect projection alignment or scene behavior in unexpected ways.

- **Keep scene and script names consistent**

Incorrect names can cause scene-loading errors or missing functionality.

- **Check in with the supervisor when unsure**

If a change may affect the experiment design, participant experience, or projection setup, discuss it with the supervisor before making permanent modifications.

- **Check that all three projectors are switched off**

Before shut down the computer, open Qlab again and select “Projector off” with the mouse and press the Spacebar and wait 5s.